Clinical Protocol I8B-MC-ITRM (a)

A Prospective, Randomized, Double-Blind Comparison of LY900014 to Insulin Lispro With an Open-Label Postprandial LY900014 Treatment Group, in Combination With Insulin Glargine or Insulin Degludec, in Adults With Type 1 Diabetes PRONTO-T1D

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PRONTO-T1D

[EudraCT number: 2015-005356-99]

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LY900014

Study I8B-MC-ITRM is a Phase 3, prospective, randomized, double-blind, outpatient, multinational, multicenter, 3-treatment group, parallel, active-controlled study conducted in patients with type 1 diabetes currently using a multiple daily injection regimen.

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Protocol Electronically Signed and Approved by Lilly: 16 Feb 2017
Amendment (a) Electronically Signed and Approved by Lilly
on approval date provided below.

Approval Date: 22-May-2017 GMT

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1. Synopsis

Title of Study:
A Prospective, Randomized, Double-Blind Comparison of LY900014 to Insulin Lispro with an Open-Label Postprandial LY900014 Treatment Group, in Combination with Insulin Glargine or Insulin Degludec, in Adults with Type 1 Diabetes PRONTO-T1D

Rationale:
Prandial insulin with faster-onset and/or faster-offset characteristics might reduce glycemic excursions and the incidence of delayed postprandial hypoglycemia compared to currently available rapid-acting insulin analogs. Rapid-acting insulins, such as Humalog®, have been shown to have a more rapid onset of action compared to human insulin; however, the general consensus is that they are not rapid enough to match carbohydrate absorption, whether delivered by pump or syringe/pen injector, limiting efficacy. An ultra-rapid-acting prandial insulin that would shift the pharmacokinetic (PK) and glucodynamic profiles of insulin to provide an even faster onset of action would better match carbohydrate absorption and allow for efficacious dosing immediately prior to meals or even after meals. Ultra-rapid insulin (URI) could be useful in the treatment of type 1 diabetes (T1D) and type 2 diabetes (T2D) in adults and children when given by multiple daily injections (MDI) or by continuous subcutaneous insulin infusion (CSII).

The aim of this study is to demonstrate that an ultra-rapid formulation of insulin lispro, LY900014, is noninferior to insulin lispro on glycemic control as measured by change from baseline to Week 26 in hemoglobin A1c (HbA1c) in patients with T1D when administered in a double-blind manner as prandial insulin in combination with basal insulin glargine or insulin degludec.

Objectives/Endpoints:

Note: Week 52 endpoints for the secondary objectives only apply to the blinded treatment arms of LY900014 and insulin lispro.

Objectives	Endpoints
Primary Objective	
1. To test the hypothesis that LY900014 is noninferior to insulin lispro on glycemic control (noninferiority margin [NIM]=0.4% for hemoglobin A1c [HbA1c]) in patients with T1D, when administered as prandial insulin (0 to 2 minutes prior to the meal), in combination with basal insulin glargine or insulin degludec for 26 weeks	1. Difference between LY900014 and insulin lispro in change from baseline to Week 26 in HbA1c
Multiplicity Adjusted Objectives	
2. To test the hypothesis that LY900014 is superior to insulin lispro in controlling 1-hour postprandial glucose (PPG) excursions, when administered as prandial insulin	2. Difference between LY900014 and insulin lispro in the 1-hour PPG excursion (serum glucose measured 1 hour after the start of the meal minus fasting serum glucose) from a mixed-meal tolerance test (MMTT) at Week 26

Multiplicity Adjusted Objectives (continued)	
3. To test the hypothesis that LY900014 is superior to insulin lispro in controlling 2-hour PPG excursions, when administered as prandial insulin	3. Difference between LY900014 and insulin lispro in the 2-hour PPG excursion (serum glucose measured 2 hours after the start of the meal minus fasting serum glucose) from an MMTT at Week 26
4. To test the hypothesis that LY900014 is superior to insulin lispro on improving glycemic control (HbA1c) when administered as prandial insulin	4. Difference between LY900014 and insulin lispro in change from baseline to Week 26 in HbA1c
5. To test the hypothesis that LY900014 administered as postprandial insulin 20 minutes after the start of a meal (LY900014+20), is noninferior to insulin lispro, administered as prandial insulin, on glycemic control (NIM=0.4% for HbA1c)	5. Difference between LY900014+20 and insulin lispro in change from baseline to Week 26 in HbA1c
Other Secondary Objectives	
6. To compare LY900014+20 with insulin lispro and LY900014 with respect to 1-hour and 2-hour PPG excursions	6. 1- and 2-hour PPG excursions from an MMTT at Week 26
7. To compare the glycemic control of LY900014 with LY900014+20	7. Change from baseline HbA1c values at Week 26
8. To compare LY900014, LY900014+20, and insulin lispro on the rate of severe hypoglycemic events	8. Rate (events/patient/100 years) of severe hypoglycemic events from baseline through Week 26 and Week 52
9. To compare LY900014, LY900014+20, and insulin lispro with respect to the incidence and rate of documented symptomatic postmeal hypoglycemia	9. Rate (events/patient/year and/or events/patient/30 days) and incidence (percent of patients with at least 1 event) of documented symptomatic postmeal hypoglycemia within 1 and 2 hours after start of a meal from Week 12 through Week 26, Week 0 through Week 26, Week 26 through Week 52, and Week 0 through Week 52
10. To compare LY900014, LY900014+20, and insulin lispro with respect to the incidence and rate of documented symptomatic hypoglycemia	10. Rate (events/patient/year and/or events/patient/30 days) and incidence (percent of patients with events) of documented symptomatic hypoglycemic events from Week 12 through Week 26, Week 0 through Week 26, Week 26 through Week 52, and Week 0 through Week 52
11. To compare LY900014, LY900014+20, and insulin lispro with respect to 1,5-Anhydroglucitol (1,5-AG)	11. Change from baseline 1,5-AG values at Week 26 and Week 52
12. To compare LY900014, LY900014+20, and insulin lispro with respect to 10-point self-monitored blood glucose (SMBG) profiles	12. Change from baseline 10-point SMBG values at Week 26 and Week 52
13. To compare LY900014, LY900014+20, and insulin lispro with respect to total, basal, and prandial insulin dose	13. Change from baseline in total, basal and prandial insulin doses and prandial/total insulin dose ratios at Week 26 and Week 52
Other Secondary Objectives (continued)	

14. To compare LY900014, LY900014+20, and insulin lispro with respect to diabetes treatment satisfaction as measured by the Insulin Treatment Satisfaction Questionnaire (ITSQ)	14. Change from baseline in ITSQ regimen inconvenience and lifestyle flexibility domain scores at Week 26 and Week 52
15. To compare LY900014, LY900014+20, and insulin lispro with respect to the proportion of patients achieving HbA1c targets	15. The proportion of patients with HbA1c <7% and ≤6.5% at Week 26 and Week 52
16. To compare the glycemic control of LY900014 and insulin lispro	16. Actual and change from baseline to Week 52 in HbA1c

Summary of Study Design:

Study I8B-MC-ITRM is a Phase 3, prospective, randomized, double-blind, outpatient, multinational, multicenter 3-treatment group, parallel, active-controlled study in patients with T1D currently using an MDI regimen.

Treatment Groups and Duration:

The study includes a 1-week screening period and an 8-week lead-in period. In 2 of the treatment groups, LY900014 and insulin lispro will be administered immediately (0-2 minutes) prior to each meal in a double-blind manner. Patients in these groups will complete a 52-week treatment period (with primary endpoint at 26 weeks), and a 4-week safety follow-up period. A third open-label treatment group consists of LY900014 administered 20 minutes after the start of a meal (LY900014+20). Patients in this treatment group will complete a 26-week treatment period and the 4-week safety follow-up period.

All patients who complete the 4-week safety follow-up visit (Visit 801) and have treatment-emergent insulin lispro antibodies that have not returned to prespecified baseline range (Visit 2) will be asked to participate in follow-up to monitor insulin lispro antibody levels. Study visits will be scheduled at approximately 3-month intervals for up to 26 weeks from Visit 801 or until insulin lispro antibodies return to baseline range, whichever occurs sooner.

Number of Patients:

Approximately 1500 participants will be screened to achieve 1199 randomized patients and 1020 patients completing 26 weeks of treatment, with approximately 646 patients completing 52 weeks.

Statistical Analysis:

The primary analysis is for the treatment period through Week 26.

Efficacy analyses will be conducted on all randomized patients according to the treatment the patients are assigned. The analyses for the primary and multiplicity adjusted objectives will be performed for the efficacy estimand including data collected prior to permanent discontinuation of investigational product (IP) through Week 26 and for the ITT estimand (treatment regimen estimand) including all data collected through Week 26 regardless of IP use. When change from baseline is included as a response variable of analysis models, the patient will be included in the analysis only if a baseline and at least 1 postbaseline measurement are available. Selected efficacy analyses will also be conducted using the Per Protocol (PP) and Completer populations. Week 52 analyses will include the LY900014 and insulin lispro treatment arms and only be performed for the efficacy estimand.

Safety analyses will be conducted on the Safety population. Analyses of adverse events (AEs) will include 2 sets of analyses, unless otherwise specified. The first set of analyses will include data collected prior to permanent

discontinuation of IP. The second set of analyses will include all data collected during the course of the entire study regardless of IP use. Analyses of hypoglycemia will be conducted on data collected prior to permanent discontinuation of IP, while analyses for post-treatment may be conducted as needed. Analyses of safety laboratory measurements will be performed on all data collected during the planned treatment period regardless of IP use. Week 26 analyses will include treatment comparisons among three treatment groups. Week 52 analyses will also include data collected during the follow-up visit for LY900014 and insulin lispro. All data collected including the follow-up visit will be summarized for LY900014+20.

Baseline is defined as the last nonmissing measurement at or before the randomization visit (Visit 8), unless otherwise specified.

For the US Food and Drug Administration (FDA) submission, the primary analysis method will use the copy reference approach to impute missing data based on multiple imputations with a pattern mixture model. This analysis is for the ITT estimand that will include all data collected from randomization through Week 26, regardless of IP use. The reference will be all observed data from the randomized patients in the same treatment arm who discontinue IP and complete the study without missing data. After imputation, the primary efficacy comparison will be based on the contrast between LY900014 and insulin lispro from the analysis of covariance (ANCOVA) analysis of change from baseline to Week 26 in HbA1c using all randomized patients. The model for this analysis will include the fixed class effects of treatment and strata (pooled country, type of basal insulin, and prandial insulin dosing plan), and the continuous, fixed covariate of baseline value.

If there are only a limited number of patients in the reference group as described above that leads to a failure in performing the proposed multiple imputation analysis, the reference will be changed to include all observed data from all randomized patients in the same treatment arm who complete the study without missing data.

For non-FDA submissions and publications, the primary efficacy comparison will be based on the contrast between LY900014 and insulin lispro at Week 26 (Visit 18) from the mixed-effect model repeated measure (MMRM) analysis of change from baseline in HbA1c including data collected from all randomized patients prior to permanent discontinuation of IP through Week 26 (efficacy estimand). The model for this analysis will include the fixed class effects of treatment and strata (pooled country, type of basal insulin, and prandial insulin dosing plan), visit, and treatment-by-visit interaction, and the continuous, fixed covariate of baseline value.

For both primary analysis approaches, LY900014 will be declared noninferior to insulin lispro if the upper limit of the 2-sided 95% confidence interval (CI) for the least squares (LS) mean difference in the change from baseline in HbA1c for LY900014 minus insulin lispro is below +0.4%. In addition, the 95% CI for the treatment difference will be compared to an alternative NIM of +0.3%. Both estimands will be tested at the full significance level of 0.05. The ANCOVA and MMRM analyses will include data from all 3 treatment groups; however, the contrasts between LY900014+20 and insulin lispro and between LY900014 and LY900014+20 are not part of the primary efficacy comparisons.

A graphical approach (Bretz et al. 2011) for multiple comparisons will be used to strongly control the overall Type I error (2-sided alpha level of 0.05) for testing the treatment effect for the primary and the following multiplicity adjusted objectives: superiority of LY900014 compared with insulin lispro for 1-hour PPG excursion at Week 26, 2-hour PPG excursion at Week 26, and change from baseline to Week 26 in HbA1c; and noninferiority of LY900014+20 to insulin lispro in change from baseline to Week 26 in HbA1c. The same graphical testing scheme will be applied for both the efficacy estimand and ITT estimand.

An analysis of covariance (ANCOVA) model with strata (pooled country, type of basal insulin, baseline HbA1c strata [(≤7.5%, >7.5%], and prandial insulin dosing plan during the study), and treatment as fixed effects and baseline as a covariate will be used to analyze the 1-hour and 2-hour PPG excursions. However, if the percentage of the patients with missing MMTT data at baseline is higher than 15%, a constrained longitudinal data analysis model (Liu et al. 2009; Lu 2010) will be used instead. Analyses details will be documented in the statistical analysis plan (SAP).

Hypoglycemia rates will be summarized for periods of 30 days, 1 year, and 100 years (severe hypoglycemia only). The rate of severe hypoglycemia per 100 years will be compared between treatment groups using the empirical method. For each of the other categories of hypoglycemia, the number of hypoglycemia events during a specific period (rate) after randomization (for example, 0-12 weeks of treatment period) will be analyzed by using a negative binomial regression model. The model will include treatment and the baseline hypoglycemia rate (measured during lead-in) as a covariate. An offset defined as the log transformation of treatment exposure in the specific period (days)/365.25 days (or 30 days) will be included in the model to estimate the rate of hypoglycemia per year (or per 30 days). The proportion of patients with at least 1 hypoglycemic event in each category (incidence) during a specific period after randomization will be analyzed using a logistic regression model including treatment and baseline hypoglycemia rate value in the model.

Continuous safety variables, as well as the change from baseline for these variables, will be analyzed by MMRM or ANCOVA models. For categorical variables, Fisher's exact test or Pearson's chi-square test will be used to compare treatment groups unless otherwise specified.

Change from baseline to last-observation-carried forward endpoints for the CCI [redacted] ITSQ, and CCI [redacted] will be analyzed using ANCOVA models.

References:

Bretz F, Posch M, Glimm E, Klinglmueller F, Maurer W, Rohmeyer K. Graphical approaches for multiple comparison procedures using weighted Bonferroni, Simes, or parametric tests. *Biom J*. 2011;53(6):894-913.

Liu GF, Lu K, Mogg R, Mallick M, Mehrotra DV. Should baseline be a covariate or dependent variable in analyses of change from baseline in clinical trials? *Stat Med*. 2009;28(20):2509-2530.

Lu K. On efficiency of constrained longitudinal data analysis versus longitudinal analysis of covariance. *Biometrics*. 2010;66(3):891-896.

2. Schedule of Activities

Study Procedure	Study Screening	Lead-In Period						Intensive Titration Period							Maintenance Period			Long Term Maintenance Period (Double Blind Arms Only)				Safety Follow-Up	ED
								Primary Treatment Period															
eCRF Visit Number	1	2	3[a]	4[a]	5	6[a]	7	8	9[a]	10	11	12[a]	13	14[a]	15	16	17	18	19	20	21	22	801	ED[b]
Visit Window (± days)		3	3	3	3	3	3	4	3	3	7	7	7	7	7	7	7	7	7	7	7	7	7	
Time on Study Relative to First Active Treatment Dose (weeks)	-9	-8	-7	-6	-4	-2	-1	0	1	2	4	6	8	10	12	18	22	26	32	39	46	52	56 (30)[c]	
Informed consent signed	X																							
IWRS	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Patient eligibility review	X																							
Randomization[d]								X																
Telephone call for visit reminder or informing patient of completion in the study																							X	
Clinical Assessments																								
Patient demographics	X																							
Medical history and preexisting conditions	X																							
Physical exam/height	X																							
Concomitant medication	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Adverse events and product complaints	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X

Study Procedure	Study Screening	Lead-In Period						Intensive Titration Period								Maintenance Period			Long Term Maintenance Period (Double Blind Arms Only)				Safety Follow-Up	ED
								Primary Treatment Period																
eCRF Visit Number	1	2	3[a]	4[a]	5	6[a]	7	8	9[a]	10	11	12[a]	13	14[a]	15	16	17	18	19	20	21	22	801	ED[b]
Visit Window (± days)		3	3	3	3	3	3	4	3	3	7	7	7	7	7	7	7	7	7	7	7	7	7	
Time on Study Relative to First Active Treatment Dose (weeks)	-9	-8	-7	-6	-4	-2	-1	0	1	2	4	6	8	10	12	18	22	26	32	39	46	52	56 (30)[c]	
Body weight[e]	X	X			X		X	X		X	X		X		X	X	X	X	X	X	X	X	X	X
Vital signs: blood pressure/pulse rate[f]	X	X			X		X	X		X	X		X		X	X	X	X	X	X	X	X	X	X
ECG[g] (12-lead; local)	X																							
Diabetes and nutrition counseling[h]		X																						
Transfer to insulin lispro[i]		X																						
Transfer to allowed study basal insulin regimen[j]		X																						
Basal and prandial insulin dose assessment[k]		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X		
Titrate basal insulin[k]		X	X	X	X	X	X																	
Titrate prandial insulin[l]								X	X	X	X	X	X	X	X									

Study Procedure	Study Screening	Lead-In Period						Intensive Titration Period								Maintenance Period			Long Term Maintenance Period (Double Blind Arms Only)				Safety Follow-Up	ED
								Primary Treatment Period																
eCRF Visit Number	1	2	3[a]	4[a]	5	6[a]	7	8	9[a]	10	11	12[a]	13	14[a]	15	16	17	18	19	20	21	22	801	ED[b]
Visit Window (± days)		3	3	3	3	3	3	4	3	3	7	7	7	7	7	7	7	7	7	7	7	7	7	
Time on Study Relative to First Active Treatment Dose (weeks)	-9	-8	-7	-6	-4	-2	-1	0	1	2	4	6	8	10	12	18	22	26	32	39	46	52	56 (30)[c]	
Ancillary Supplies/Diaries/IP																								
Dispense blood glucose meter and ancillary supplies and complete training[m,n]		X						X		X	X		X		X	X	X	X	X	X	X	X	X	
Dispense eCOA and complete training[n]		X																						
Train on collecting 4- and 10-point SMBG profiles[n]		X																						
Review 4-point SMBG profiles			X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Remind patient of 10-point SMBG requirements[o]						X	X			X		X		X			X		X		X			
Review 10-point SMBG Profiles								X			X		X		X			X		X		X		X[p]
Review/discuss hypo data			X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X		X
Dispense IP		X			X			X			X		X		X	X	X	X	X	X	X			
Return of eCOA devices																							X	X

Study Procedure	Study Screening	Lead-In Period						Intensive Titration Period								Maintenance Period			Long Term Maintenance Period (Double Blind Arms Only)				Safety Follow-Up	ED
								Primary Treatment Period																
eCRF Visit Number	1	2	3[a]	4[a]	5	6[a]	7	8	9[a]	10	11	12[a]	13	14[a]	15	16	17	18	19	20	21	22	801	ED[b]
Visit Window (± days)		3	3	3	3	3	3	4	3	3	7	7	7	7	7	7	7	7	7	7	7	7	7	
Time on Study Relative to First Active Treatment Dose (weeks)	-9	-8	-7	-6	-4	-2	-1	0	1	2	4	6	8	10	12	18	22	26	32	39	46	52	56 (30)[c]	
Patient returns used and unused study drug supplies					X			X			X		X		X	X	X	X	X	X	X	X		X
Drug accountability					X			X			X		X		X	X	X	X	X	X	X	X		X
Laboratory Assessments																								
Patient fasts prior to visit		X						X										X				X		X
MMTT[q]								X										X						
Urinalysis panel	X																							
Pregnancy test[r]	X							X																
Follicle-stimulating hormone test[s]	X																							
Chemistry panel	X							X										X				X		X
Fasting Serum Glucose		X																						
Hematology	X							X										X				X		X
1,5-Anhydroglucitol								X			X				X			X		X		X		X
Hemoglobin A1c	X						X	X			X		X		X			X		X		X	X	X
Lipid profile								X										X				X		X
Pharmacogenetic sample[t]								X																

Study Procedure	Study Screening	Lead-In Period						Intensive Titration Period								Maintenance Period			Long Term Maintenance Period (Double Blind Arms Only)				Safety Follow-Up	ED
								Primary Treatment Period																
eCRF Visit Number	1	2	3[a]	4[a]	5	6[a]	7	8	9[a]	10	11	12[a]	13	14[a]	15	16	17	18	19	20	21	22	801	ED[b]
Visit Window (± days)		3	3	3	3	3	3	4	3	3	7	7	7	7	7	7	7	7	7	7	7	7	7	
Time on Study Relative to First Active Treatment Dose (weeks)	-9	-8	-7	-6	-4	-2	-1	0	1	2	4	6	8	10	12	18	22	26	32	39	46	52	56 (30)[c]	
Nonpharmacogenetic biomarker storage samples								X										X						X
Anti-insulin lispro antibodies		X						X		X	X				X			X				X	X[u]	X
Health Outcomes Questionnaires[v]																								
ITSQ		X						X										X				X		X
CCI		X						X										X				X		X
		X						X										X				X		X

Schedule of Activities

Abbreviations: ECG = electrocardiogram; eCOA = electronic clinical outcomes assessment, eCRF = electronic case report form; ED = early discontinuation; CCI glucose; hypo = hypoglycemia; IP = investigational product; ITSQ = Insulin Treatment Satisfaction Questionnaire; IWRS = interactive web response system; MMTT = mixed-meal tolerance test; SMBG = self-monitored blood glucose; V = visit; CCI

[a] Telephone visits are indicated by shaded columns. Activities include:
- Record visit in the IWRS.
- Collect adverse events and concomitant medications.
- Review blood glucose readings and study drug doses.
- Adjust prandial insulin or basal insulin doses as needed.
- Review hypoglycemic events.
- Provide reminders regarding scheduled visits, fasting, and SMBG profiles, as applicable.

[b] Patients who have been randomized will be asked to return for the ED visit in a fasting state unless patient is already fasting and at the site when the decision to discontinue is made. Patients who discontinue during the lead-in period prior to randomization who are not already at the site, will be asked to return for the ED in a nonfasting state and all activities should be completed except for laboratory tests and questionnaires (ITSQ, CCI).

[c] Patients randomized to the open-label treatment group will complete the study at Week 30 unless patient continues into the insulin antibody assessment period. It is recommended that the first insulin lispro antibody follow-up visit (Visit 802 [Appendix 8]) be scheduled for all study patients, regardless of treatment-emergent antibody status. Visit 802 should occur approximately 13 weeks after Visit 801.

[d] Randomization should occur after all Visit 8 procedures including MMTT. If MMTT is rescheduled post Visit 8, randomization should not occur until baseline MMTT is completed. The patient will administer their first dose of study insulin with the first meal after the MMTT has been completed and randomization has occurred.

[e] Patients should be advised to remove their shoes and empty their pockets before the body weight is obtained.

[f] Vital sign measurements must be taken before obtaining an ECG tracing and before collection of blood samples for laboratory testing. These measurements should be determined after patients have been seated quietly for at least 5 minutes in a chair with feet on the floor. The arm used for blood pressure measurement should be supported at heart level.

[g] Patients must be supine for approximately 5 to 10 minutes before ECG collection and remain supine but awake during ECG collection.

[h] Initial training at Visit 2 will include diabetes education and nutrition counseling. Appropriate site personnel will administer training and education using locally approved diabetes education/training materials or other materials provided by the Sponsor. Patients may be provided abbreviated training and education at visits following Visit 2 as appropriate.

[i] Only applies to patients on insulin glulisine or insulin aspart.

[j] Patients will be transferred to a study basal insulin regimen at Visit 2. Investigators will determine the appropriate basal insulin regimen for each patient. It is expected that most of the basal insulin dose adjustments should occur during the lead-in period. However, basal insulin titration may continue during the intensive titration period for patients who need more intensification to achieve glycemic targets. Basal insulin dose adjustments during the maintenance period should be made for safety reasons such as hypoglycemia and unacceptable hyperglycemia.

[k] Assessments of the basal insulin dose should be made, at minimum, weekly during the lead-in period, including Weeks -5 and -3. Assessment of the prandial insulin dose should be made, at minimum, weekly during the initial 12 weeks after randomization, including Weeks 3, 5, 7, 9, and 11.

[l] During the lead-in period, prandial doses, correction factors, and insulin to carbohydrate ratios should be evaluated and adjustments should be made for safety reasons such as hypoglycemia and unacceptable hyperglycemia. During the intensive titration period, prandial insulin dose should be adjusted in order to meet the target SMBG levels. Further adjustments during the maintenance period should be made for safety reasons such as hypoglycemia and unacceptable hyperglycemia.

[m] Glucose monitoring supplies will be dispensed at other visits, as needed.

[n] Training may be repeated at other visits, as needed.

[o] Patients in all 3 study arms should be instructed to perform three 10-point SMBG profiles during a 2-week period. The 10-point profile is completed over a 1-day period, preferably on 3 nonconsecutive days (weekdays and weekends). Ten-point SMBG profiles should not be performed on the day of MMTT.

[p] Review of 10-point SMBG profiles at the ED visit will take place only for patients randomized into the study.

[q] MMTT can occur 0 to 4 days prior to the visit, and patient must be fasting prior to start of the MMTT.

[r] Serum pregnancy test must be performed in women of childbearing potential at Visit 1 followed by a urine pregnancy test within 24 hours prior to IP exposure at randomization (Visit 8) and at other times at the investigator's discretion. When required per local regulations and/or institutional guidelines, local pregnancy testing will occur at mandatory times during the study treatment period.

[s] Follicle-stimulating hormone test must be performed at Visit 1 for a postmenopausal woman who is between 50 and 54 years of age (inclusive) with an intact uterus, not on hormone therapy, and who has had at least 6 months of spontaneous amenorrhea.

[t] Pharmacogenetic sample can be collected at any other visit if not collected at Visit 8

[u] Patients with treatment-emergent insulin lispro antibodies measured at Visit 801 that have not returned to prespecified baseline range (Visit 2), will be asked to undergo additional antibody monitoring as described in Appendix 8. This follow-up assessment also applies to any patient who discontinues study insulin treatment prior to the end of the planned treatment period who has a treatment-emergent antibody response that has not returned to baseline range by Visit 801. It is recommended that the first insulin lispro antibody follow-up visit (Visit 802) be scheduled for all study patients, regardless of treatment-emergent antibody status.

[v] Health outcomes questionnaires will be administered at the study sites based on the availability of appropriate translations. The questionnaires should be administered prior to other study procedures except at Visit 8 and Visit 18 when questionnaires may be administered following the start of the MMTT.

3. Introduction

3.1. Study Rationale

Prandial insulin with faster-onset and/or faster-offset characteristics might reduce glycemic excursions and decrease the incidence of delayed postprandial hypoglycemia compared to currently available rapid-acting insulin analogs. The insulin analog, insulin lispro (Humalog®), has been shown to be absorbed more quickly than regular human insulin (Humalog package insert, 2015). In healthy volunteers given subcutaneous (SC) doses of insulin lispro ranging from 0.1 to 0.4 U/kg, peak serum levels were seen 30 to 90 minutes after dosing. Rapid-acting insulins have been shown to have a more rapid onset of action compared to human insulin; however, the general consensus is that they are not rapid enough to match carbohydrate absorption and many patients are unable to achieve optimal glycemic control. An ultra-rapid-acting prandial insulin with pharmacokinetic (PK) and glucodynamic (GD) profiles that demonstrate faster absorption and onset of action may better match carbohydrate absorption and lead to improved postprandial control. The time action profile of rapid-acting insulin could be enhanced through the addition of excipients to an existing formulation to increase capillary blood flow and/or enhance vascular permeability. An ultra-rapid insulin (URI) would be useful in the treatment of type 1 diabetes (T1D) and type 2 diabetes (T2D) when delivered by multiple daily injections (MDI), by continuous SC insulin infusion (CSII), and in the development of closed loop insulin delivery systems.

The aim of this study is to demonstrate that an ultra-rapid formulation of insulin lispro, LY900014, is noninferior to insulin lispro on glycemic control as measured by change from baseline to Week 26 in hemoglobin A1c (HbA1c) in patients with T1D, when administered in a double-blind manner as prandial insulin in combination with basal insulin glargine or insulin degludec.

3.2. Background

T1D is an autoimmune disease mediated by a combination of genetic and environmental triggers resulting in lymphocytic infiltration of pancreatic islets, destruction of beta cells, and lifelong dependency on exogenous insulin. T1D prevalence is near 1 in 300 and its incidence is steadily increasing worldwide (3% per year [Gan et al. 2012]).

There have been many advances in the treatment of T1D in the last 20 years; however, reaching and maintaining glycemic goals remains challenging even under intensive insulin therapy regimens. Only approximately 30% of insulin requiring diabetes patients are able to reach the goal HbA1c target of <7% (Stark-Casagrande et al. 2013). Currently available rapid-acting insulin analogs continue to be unable to match the kinetics of physiological postmeal insulin secretion, which is biphasic. Normal first phase insulin response is a rapid but short-lived increase in secretion, which is then followed by a more slowly developing prolonged increase (Barrett et al. 2016). First phase insulin release prevents the rapid development of postprandial hyperglycemia while second phase release of insulin ensures that glucose enters tissues in a steady, controlled manner throughout the late postprandial period. Thus, there remains a need to

continue to develop formulations with a time action profile that more closely approximates that of endogenous insulin section.

Ideally, currently available rapid-acting insulin analogs should be injected 10 to 15 minutes prior to meal consumption in order to control postprandial glucose (PPG). However, many persons inject their rapid-acting insulin at the time of the meal or after the meal. According to survey data from the T1D Exchange on the timing of prandial insulin injection, 16.3% of patients indicated that they inject 15 minutes prior to meals, 33% inject 1 to 14 minutes prior to meals, 38.6% inject at the time of the meal, and 12.1% inject after meals. Because patients often inject later than recommended there is a greater mismatch between insulin action and postprandial blood glucose (BG) elevations. With postmeal dosing, the mismatch between the rise in BG and the onset of insulin action is even more pronounced. For the development of a URI, it will be important to understand the relationship between time action profile of the insulin, insulin injection timing, and meal timing in order to maximize improvements in postprandial glycemic control and minimize hypoglycemia risk. A URI with higher early insulin concentration and peak exposure, and shorter duration should improve early postprandial control and limit postmeal hyperglycemia, while reducing late postprandial hypoglycemia due to lower insulin exposure.

HbA1c provides an integrated measurement of both fasting and postprandial glycemic control and is the most reliable marker for overall glucose exposure. Elevation in HbA1c is the best predictor of diabetes complications. Control of both fasting and postprandial hyperglycemia is essential to reach HbA1c goals. The relative contribution of postprandial hyperglycemia is predominant with moderate to fairly well controlled HbA1c levels (Monnier et al. 2003).

A new pharmaceutical innovation that may allow more effective control of PPG levels is LY900014, a new formulation of insulin lispro developed as an ultra-rapid acting insulin with a faster onset of action and shorter duration of action compared to currently available rapid-acting insulin analogs. The changes in PK and GD characteristics are achieved by coformulating insulin lispro with treprostinil and ingredients Generally Recognized As Safe (GRAS) by the Food and Drug Administration (FDA) as excipients.

LY900014 is a formulation of insulin lispro that contains the prostacyclin analog treprostinil, citrate, and other excipients. Treprostinil as an excipient enhances the absorption of insulin lispro by local vasodilatation rather than as an active pharmaceutical ingredient that elicits a systemic effect. Treprostinil is a prostacyclin analog administered either by inhalation (Tyvaso®), intravenous (IV) infusion, or continuous SC infusion (Remodulin®) for the treatment of symptomatic pulmonary arterial hypertension, and has been approved in the United States since 2002 (Remodulin package insert, 2014) and in Europe since 2005 (PMR [WWW]). Sodium citrate, an excipient that speeds insulin absorption, is also included in the formulation to further enhance the absorption of insulin lispro. Sodium citrate and the other excipients in the LY900014 formulation are listed in the FDA GRAS food additives database and in the FDA Inactive Ingredients in Approved Drugs database. Furthermore, the excipient concentration in LY900014 is within the limits identified for approved drug products in the FDA Inactive Ingredients in Approved Drugs database.

Please refer to the Humalog local product labeling (for example, Humalog package insert, 2015; Humalog Summary of Product Characteristics, 2016) for more information regarding insulin lispro.

The Investigator's Brochure (IB) describes the clinical and nonclinical development of LY900014.

3.3. Benefit/Risk Assessment

More detailed information about the known and expected benefits and risks of insulin lispro may be found in the country-specific product labeling (for example, Patient Information Leaflet, Package Insert, or Summary of Product Characteristics).

Across all doses in the Eli Lilly and Company (Lilly) clinical studies that have evaluated treprostinil as a local vasodilator with or without insulin lispro, there was no clinically significant increase in those adverse events (AEs) associated with systemic absorption of treprostinil, as described in the Remodulin package insert (2014) (that is, headache, diarrhea, nausea, jaw pain, vasodilatation, rash, edema, anorexia, vomiting, asthenia, abdominal pain, and hypotension). The exposures of treprostinil in LY900014 for participants in upcoming and future clinical trials are expected to be much lower than those observed in the dose ranges previously explored with SC bolus administration of treprostinil and are expected to be substantially lower than those observed in the treatment of pulmonary artery hypertension (PAH).

More information about the known and expected benefits, risks, serious AEs (SAEs), and reasonably anticipated AEs of LY900014 can be found in the IB.

4. Objectives and Endpoints

Table ITRM.1 shows the objectives and endpoints of the study. Week 52 endpoints for the secondary objectives only apply to the blinded treatment arms of LY900014 and insulin lispro.

Table ITRM.1. Objectives and Endpoints

Objectives	Endpoints
Primary Objective	
1. To test the hypothesis that LY900014 is noninferior to insulin lispro on glycemic control (NIM=0.4% for HbA1c) in patients with T1D, when administered as prandial insulin (0 to 2 minutes prior to the meal), in combination with basal insulin glargine or insulin degludec for 26 weeks	1. Difference between LY900014 and insulin lispro in change from baseline to Week 26 in HbA1c
Multiplicity Adjusted Objectives	
2. To test the hypothesis that LY900014 is superior to insulin lispro in controlling 1-hour PPG excursions, when administered as prandial insulin	2. Difference between LY900014 and insulin lispro in the 1-hour PPG excursion (serum glucose measured 1 hour after the start of the meal minus fasting serum glucose) from a MMTT at Week 26
3. To test the hypothesis that LY900014 is superior to insulin lispro in controlling 2-hour PPG excursions, when administered as prandial insulin	3. Difference between LY900014 and insulin lispro in the 2-hour PPG excursion (serum glucose measured 2 hours after the start of the meal minus fasting serum glucose) from an MMTT at Week 26
4. To test the hypothesis that LY900014 is superior to insulin lispro on improving glycemic control (HbA1c) when administered as prandial insulin	4. Difference between LY900014 and insulin lispro in change from baseline to Week 26 in HbA1c
5. To test the hypothesis that LY900014 administered as postprandial insulin 20 minutes after the start of a meal, (LY900014+20), is noninferior to insulin lispro, administered as prandial insulin, on glycemic control (NIM=0.4% for HbA1c)	5. Difference between LY900014+20 and insulin lispro in change from baseline to Week 26 in HbA1c
Other Secondary Objectives	
6. To compare LY900014+20 with insulin lispro and LY900014 with respect to 1-hour and 2-hour PPG excursions	6. 1- and 2-hour PPG excursions from an MMTT test at Week 26
7. To compare the glycemic control of LY900014 with LY900014+20	7. Change from baseline HbA1c values at Week 26
8. To compare LY900014, LY900014+20, and insulin lispro on the rate of severe hypoglycemic events	8. Rate (events/patient/100 years) of severe hypoglycemic events from baseline through Week 26 and Week 52

Other Secondary Objectives (continued)	
9. To compare LY900014, LY900014+20, and insulin lispro with respect to the incidence and rate of documented symptomatic postmeal hypoglycemia	9. Rate (events/patient/year and/or events/patient/30 days) and incidence (percent of patients with at least 1 event) of documented symptomatic postmeal hypoglycemia within 1 and 2 hours after start of a meal from Week 12 through Week 26, Week 0 through Week 26, Week 26 through Week 52, and Week 0 through Week 52
10. To compare LY900014, LY900014+20, and insulin lispro with respect to the incidence and rate of documented symptomatic hypoglycemia	10. Rate (events/patient/year and/or events/patient/30 days) and incidence (percent of patients with events) of documented symptomatic hypoglycemic events from Week 12 through Week 26, Week 0 through Week 26, Week 26 through Week 52, and Week 0 through Week 52
11. To compare LY900014, LY900014+20, and insulin lispro with respect to 1,5-AG	11. Change from baseline 1,5-AG values at Week 26 and Week 52
12. To compare LY900014, LY900014+20, and insulin lispro with respect to 10-point SMBG profiles	12. Change from baseline 10-point SMBG values at Week 26 and Week 52
13. To compare LY900014 , LY900014+20, and insulin lispro with respect to total, basal, and prandial insulin dose	13. Change from baseline in total, basal and prandial insulin doses and prandial/total insulin dose ratios at Week 26 and Week 52
14. To compare LY900014, LY900014+20, and insulin lispro with respect to diabetes treatment satisfaction as measured by the ITSQ	14. Change from baseline in ITSQ regimen inconvenience and lifestyle flexibility domain scores at Week 26 and Week 52
15. To compare LY900014, LY900014+20, and insulin lispro with respect to the proportion of patients achieving HbA1c targets	15. The proportion of patients with HbA1c <7% and ≤6.5% at Week 26 and Week 52
16. To compare the glycemic control of LY900014 and insulin lispro	16. Actual and change from baseline to Week 52 in HbA1c
Tertiary/Exploratory Objectives	
17. To compare the safety of LY900014, LY900014+20, and insulin lispro	17. Adverse events, vital signs, chemistry, and hematology laboratory measures
18. To compare the incidence of treatment-emergent positive anti-insulin lispro antibodies for LY900014, LY900014+20, and insulin lispro	18. Incidence of treatment-emergent positive anti-insulin lispro antibodies
CCI	
21. To compare LY900014, LY900014+20, and insulin lispro with respect to changes in body weight	21. Change in weight (kg) from baseline to Week 26 and Week 52

Tertiary/Exploratory Objectives (continued)	
22. To compare LY900014, LY900014+20, and insulin lispro with respect to the proportion of patients achieving improvement from baseline in HbA1c targets	22. The proportions of patients with shifts in HbA1c to <8% and ≤9%, and >9% from baseline to Week 26 and Week 52
23. To compare LY900014, LY900014 +20, and insulin lispro with respect to glycemic variability	23. Within-day and between-day glycemic variability measured by the standard deviation and the coefficient of variation of 10-point SMBG profiles

Abbreviations: 1,5-AG = 1,5-Anhydroglucitol; CCI [redacted] HbA1c = hemoglobin A1c; ITSQ = Insulin Treatment Satisfaction Questionnaire; MMTT = mixed-meal tolerance test; NIM = noninferiority margin; PPG = postprandial glucose; SMBG = self-monitored blood glucose; T1D = type 1 diabetes mellitus; CCI [redacted]

5. Study Design

5.1. Overall Design

Study I8B-MC-ITRM (ITRM) is a Phase 3, prospective, randomized, outpatient, multinational, multicenter, 3-treatment group, parallel, active-controlled study conducted in patients with T1D currently using an MDI regimen. In 2 of the treatment groups, LY900014 and insulin lispro will be administered immediately (0-2 minutes) prior to each meal in a double-blind manner. A third open-label treatment group consists of LY900014 administered 20 minutes after the start of a meal (LY900014+20). It is not possible to blind this treatment group with different injection timing. The study is designed to demonstrate noninferiority of LY900014 when compared with insulin lispro in change from baseline to Week 26 in HbA1c, when both are administered at the start of the meal. For patients in the 2 double-blind arms, the study includes a 1-week screening period and an 8-week lead-in period followed by a 52-week treatment period and a 4-week safety follow-up period. For patients in the open-label treatment group, the treatment period will end after 26 weeks, which will be followed by a 4-week safety follow-up period.

All patients who complete the 4-week safety follow-up visit (Visit 801) and have treatment-emergent insulin lispro antibodies that have not returned to the prespecified baseline range (Visit 2) will be asked to participate in follow-up to monitor insulin lispro antibody levels for up to 26 weeks after Visit 801 as described in Appendix 8.

The purpose of the lead-in period (prior to randomization) will be to titrate basal insulin, obtain preliminary diagnostic laboratory tests, and determine baseline hypoglycemia rates. Patients treated with insulin glargine U-100 (Basaglar/Abasaglar or LANTUS) once or twice daily, insulin glargine U-300 once daily, insulin degludec U-100 or U-200 once daily, insulin detemir U-100 once or twice daily, or neutral protamine Hagedorn (NPH) once or twice daily will be eligible for inclusion in the trial. At Visit 2, patients will be switched to an allowed study basal insulin regimen; insulin glargine U-100 twice daily, insulin glargine U-100 once daily (either in the morning or evening), or insulin degludec U-100 once daily as determined by the investigator. The brand of insulin glargine chosen at Visit 2 should remain the same throughout the study; switching brands should be avoided. Insulin glargine or insulin degludec will be titrated during the 8-week lead-in period using a titration algorithm that should enable patients to reach the target fasting blood glucose (FBG) by the end of this period.

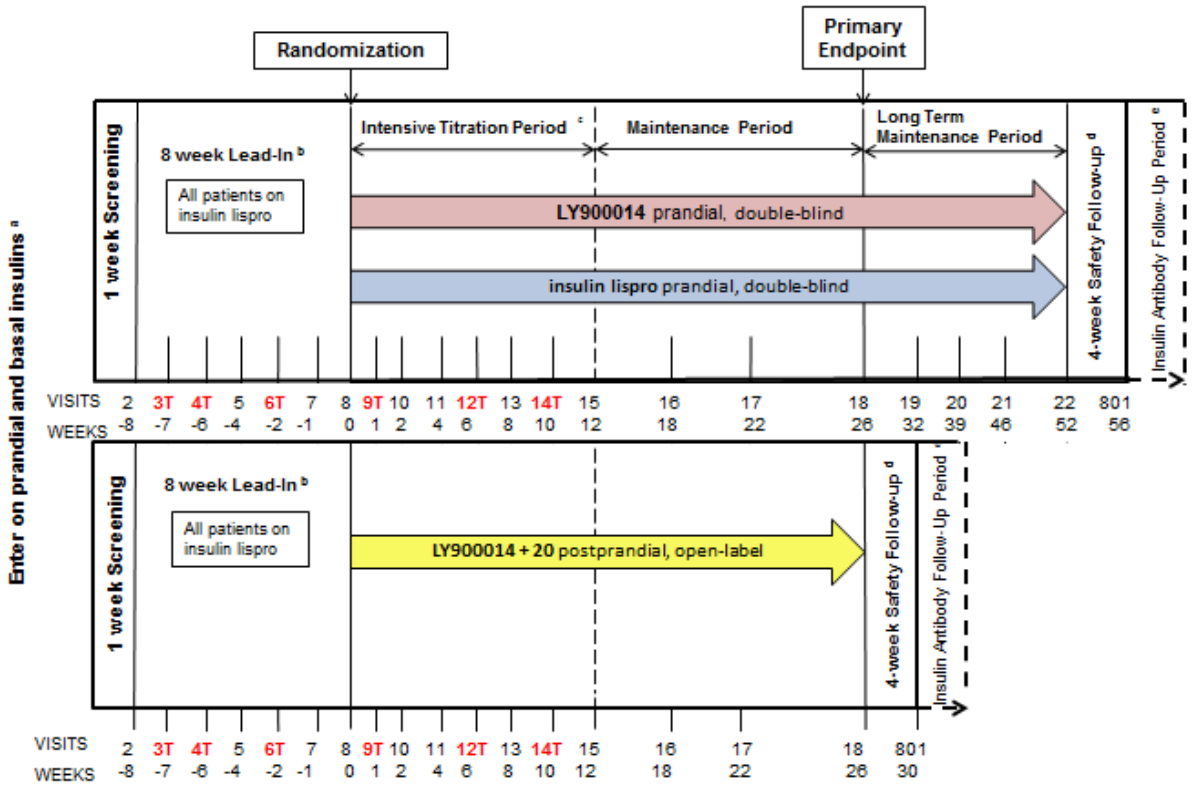
All patients using either insulin aspart or insulin glulisine at screening will be transferred to insulin lispro at Visit 2 so that all patients are using insulin lispro during the lead-in period. Prandial insulin doses or insulin to carbohydrate ratio (ICR), and correction factor (CF) should not be changed during the lead-in period except for safety reasons (hypoglycemia or unacceptable hyperglycemia) or to facilitate basal insulin optimization.

At Visit 8, patients will be randomized to either LY900014 at mealtime, insulin lispro at meal-time, or LY900014 administered 20 minutes after the start of a meal (LY900014+20). During the initial 12 weeks after randomization (intensive titration period), prandial insulin dose should be adjusted as necessary in order to meet the target self-monitored blood glucose (SMBG) levels. Basal insulin may be titrated as needed to facilitate optimal prandial dosing or for safety reasons

such as hypoglycemia or unacceptable hyperglycemia. Thereafter, during the maintenance period (Weeks 12-26 of treatment), it is expected that adjustments to prandial and basal insulin doses would be to maintain glycemic control or for safety reasons such as hypoglycemia or unacceptable hyperglycemia.

For patients continuing into the 26-week long term maintenance period (2 double-blind arms), adjustments to both prandial and basal insulins should be made in order to maintain or further optimize glycemic control, as appropriate.

Figure ITRM.1 illustrates the study design.



Abbreviation: T = telephone visit.

a At Visit 2, patients on insulin glulisine or insulin aspart will be transferred to insulin lispro. The patients' basal insulin regimen will be switched to insulin glargine U-100 (once or twice daily) or to insulin degludec U-100 once daily. At Visit 8, patients will be randomized to either premeal insulin lispro, premeal LY900014, or LY900014+20 and continue their basal insulin regimen.

b Titrate basal insulin (see Section 7.2.1.3).

c Titrate prandial insulin (insulin lispro or LY900014) (see Section 7.2.1.4).

d Patients will discontinue study insulins at Week 26 for the LY900014+20 open-label treatment group or Week 52 for the 2 blinded treatment groups.

e Eligible patients will have visits at approximately 3-month intervals for up to 26 weeks after Visit 801 for follow-up of insulin lispro antibody levels (see Appendix 8).

Figure ITRM.1. Illustration of study design for Clinical Protocol I8B-MC-ITRM.

5.2. Number of Participants

Approximately 1500 participants will be screened to achieve 1199 randomized patients and 1020 patients completing 26 weeks and approximately 646 patients completing 52 weeks (double-blind arms only).

5.3. End of Study Definition

End of the study is the last scheduled procedure shown in the Insulin Lispro Antibody Follow-Up Assessment Schedule of Activities (Appendix 8) for the last patient.

5.3.1. Safety Follow-Up

Safety follow-up visit guidelines are as follows (see Section 2):

- Patients who discontinue from the study early during the lead-in period (prior to randomization) should complete an early discontinuation visit only.
- Patients who discontinue from investigational product (IP) early but remain in the study should complete all remaining visits per Schedule of Activities, including Visit 801.
- Patients who discontinue from the study early (regardless of whether they discontinue IP at the same time or have discontinued IP at an earlier visit) should complete an early discontinuation visit followed by the safety follow-up visit (Visit 801) per Schedule of Activities.
- Patients in the 2 double-blind arms who finish Visit 22 without early discontinuation of IP should complete a safety follow-up visit (Visit 801) 4 weeks after Visit 22.
- Patients in the open-label treatment group who finish Visit 18 without early discontinuation of IP should complete a safety follow-up visit (Visit 801) 4 weeks after Visit 18.

All patients who complete the 4-week safety follow-up visit (Visit 801) and have treatment-emergent insulin lispro antibodies that have not returned to prespecified baseline range (Visit 2) will be asked to participate in follow-up to monitor insulin lispro antibody levels as described in Appendix 8. Patients participating in the antibody follow-up period will complete the study when insulin antibodies have returned to baseline range or when Visit 803 is completed, whichever occurs first.

5.4. Scientific Rationale for Study Design

Study ITRM is a Phase 3 study to evaluate LY900014 compared to insulin lispro each in combination with basal insulin in patients with T1D. The trial has 3 treatment groups, 2 of which are double-blind to allow comparison of LY900014 and insulin lispro when injected at the start of the meal. The third treatment group is open-label due to inability to blind postprandial injection timing of LY900014 (20 minutes after the start of a meal) and will evaluate the efficacy and safety of postprandial dosing. Patients in the 2 blinded arms will continue with double-blind treatment for an additional 26 weeks for a total of 52 weeks to allow comparison of long-term safety and efficacy in a blinded manner. Patients in the open-label treatment group will complete their study participation after the 26-week primary endpoint and safety follow-up.

The study is designed to compare HbA1c lowering as the primary endpoint, a measure of glycemic control accepted by health care providers and regulatory authorities as a validated measure of BG control over time and is the best marker for development and progression of diabetes complications. The lead-in period consists of 8 weeks to allow for optimization of basal insulin dosing. The 26-week treatment period consists of a 12-week intensive titration phase to optimize prandial insulin dosing followed by a 14-week maintenance phase, so the primary endpoint HbA1c reflects glucose control during the 3-month time period when there are minimal changes to insulin dose. The second 26-week period is a long-term maintenance phase to demonstrate durability of glycemic control.

5.5. Justification for Dose

LY900014 will have the same insulin lispro concentration (100 U/mL) as that of commercially available Humalog. The addition of treprostinil to the insulin lispro formulation does not modify the physical, chemical, or biological integrity of insulin lispro. The dosage of basal and prandial insulins used in this study should be determined based on the individual needs of each patient.

6. Study Population

This study will include patients who have been diagnosed with T1D for at least 1 year, using insulin continuously for at least 1 year, currently on MDI therapy for at least 90 days, and having an HbA1c value of ≥7.0 and ≤9.5% at screening. Patients' basal insulin may be insulin glargine U-100 (Basaglar/Abasaglar or LANTUS) once or twice daily, insulin glargine U-300 once daily, insulin degludec U-100 or U-200 once daily, insulin detemir U-100 once or twice daily, or neutral protamine Hagedorn (NPH) once or twice daily. Patients may be on any approved injectable rapid-acting insulin analog.

Patients must give written informed consent (approved by Lilly or its designee and the ethical review board [ERB] governing the site) before being allowed to participate in the study and before any screening assessments are performed.

Study investigator(s) will review the patient's records and/or history and screening test results/measurements to determine if the patient meets all inclusion and no exclusion criteria to qualify for participation in the study. All screening activities must be completed and reviewed before the patient begins the lead-in period.

Prospective approval of protocol deviations to recruitment and enrollment criteria, also known as protocol waivers or exemptions, are not permitted.

6.1. Inclusion Criteria

Patients are eligible to be included in the study only if they meet all of the following criteria at screening:

Type of Patient and Disease Characteristics

[1] Men or women diagnosed (clinically) with T1D, based on the World Health Organization (WHO) classification (Appendix 5) for at least 1 year prior to screening, and continuously using insulin for at least 1 year

Patient Characteristics

[2] Are at least 18 years of age

[3] Have been on MDI therapy including a rapid acting insulin analog (insulin lispro U-100, insulin aspart, insulin glulisine) for at least 90 days

[4] Have been treated for at least 30 days prior to screening with one of the following:

a) Insulin glargine U-100 or U-300

b) Insulin detemir U-100

c) Insulin degludec U-100 or U-200

d) NPH

[5] Have an HbA1c value ≥7.0 and ≤9.5%, according to the central laboratory at screening (Visit 1).

[6] Have a body mass index (BMI) of ≤35.0 kg/m^2 at screening (Visit 1).

[7] Male patients:

a) No male contraception required except in compliance with specific local government study

[8] Female patients:

a) Women not of childbearing potential may participate and include those who are:

i) infertile due to surgical sterilization (hysterectomy, bilateral oophorectomy, or tubal ligation), congenital anomaly such as Mullerian agenesis;

Or

ii) postmenopausal – defined as either

(1) a woman 50 to 54 years of age (inclusive) with an intact uterus, not on hormone therapy who has had either

(a) cessation of menses for at least 1 year;

Or

(b) at least 6 months of spontaneous amenorrhea with a follicle-stimulating hormone >40 mIU/mL;

Or

(2) a woman 55 or older not on hormone therapy, who has had at least 6 months of spontaneous amenorrhea;

Or

(3) a woman at least 55 years of age with a diagnosis of menopause prior to starting hormone replacement therapy

b) Women of childbearing potential participating:

i) Cannot be pregnant or intend to become pregnant

ii) Cannot be breastfeeding (including the use of a breast pump)

iii) Must remain abstinent or use 1 highly effective method of contraception or a combination of 2 effective methods of contraception for the entirety of the study (See Appendix 7)

iv) Test negative for pregnancy at the time of screening (Visit 1). Note: a urine pregnancy test is conducted at Visit 8.

[9] Have access to a telephone or alternative means for close monitoring and communications, and have access to a reliable cellular signal for transmission of the electronic clinical outcomes assessment (eCOA) data

[10] Have refrigeration in the home or have ready access to refrigeration for storage of insulin therapy

[11] Patient whom the investigator has determined can be randomized and maintain the treatment regimens based on their previous medical history including insulin dosing regimens, hypoglycemic episodes, and glycemic control

[12] Capable of, willing, and desirous to do the following:

a) Inject insulin with the use of an insulin injection device (insulin pen) according to included directions

b) Perform self-BG monitoring including 10-point SMBG on designated days (patients using a personal continuous glucose monitoring [CGM] or flash glucose monitoring [FGM] device for insulin dosing decisions must still use the study glucose meter according to the protocol)

c) Keep records in eCOA as required by this protocol

d) Participate in two 4-hour mixed-meal tolerance tests (MMTTs) and consume a standardized meal for the tests

e) Follow an algorithm for basal insulin adjustment and individualized prandial insulin dosing using fixed pattern adjustment or carbohydrate counting as agreed upon with the investigator

f) Comply with the use of the study insulin and scheduled visits

[13] Considered healthy (apart from T1D) upon completion of medical history, physical examination, vital signs, electrocardiogram (ECG), and analysis of laboratory safety variables, as judged by the investigator

Informed Consent

[14] Have given written informed consent to participate in this study in accordance with local regulations.

6.2. Exclusion Criteria

Patients will be excluded from study enrollment if they meet any of the following criteria at screening:

Medical Conditions

[15] Have any other condition (including known drug or alcohol abuse, or psychiatric disorder including eating disorder) that precludes the patient from following and completing the protocol

[16] Have hypoglycemia unawareness as judged by the investigator

[17] Have had more than 1 episode of severe hypoglycemia (defined as requiring assistance due to neurologically disabling hypoglycemia) within the last 6 months prior to screening

[18] Have had more than 1 emergency room visit or hospitalization due to poor glucose control (hyperglycemia or diabetic ketoacidosis) within 6 months prior to screening (Visit 1)

[19] Have cardiovascular disease within the last 6 months prior to screening, defined as stroke, decompensated heart failure New York Heart Association class III or IV (see Appendix 6), myocardial infarction, unstable angina pectoris, or coronary arterial bypass graft

[20] Renal:

a) History of renal transplantation

b) Currently receiving renal dialysis

c) Serum creatinine >2.0 mg/dL (177 µmol/L) at screening

[21] Hepatic: Have obvious clinical signs or symptoms of liver disease (for example, acute or chronic hepatitis, or cirrhosis) or elevated liver enzyme measurements as indicated below at screening (Visit 1):

a) Total bilirubin level (TBL) ≥2X the upper limit of normal (ULN) (with the exception of Gilberts Disease) as defined by the central laboratory,

OR

b) Alanine aminotransferase (ALT) ≥3X ULN as defined by the central laboratory

OR

c) Aspartate aminotransferase (AST) ≥3X ULN as defined by the central laboratory

[22] Malignancy: Have active or untreated malignancy, have been in remission from clinically significant malignancy (other than basal cell or squamous cell skin cancer) for less than 5 years, or are at an increased risk for developing cancer or a recurrence of cancer in the opinion of the investigator

[23] Have any hypersensitivity or allergy to any of the insulins or excipients used in this trial

[24] Have hypersensitivity or allergy to any of the ingredients in the standardized test meal (MMTT) (for example, nut allergy)

[25] Hematologic: Have had a blood transfusion or severe blood loss within 90 days prior to screening or have known hemoglobinopathy, anemia, or any other traits known to interfere with measurement of HbA1c

[26] Have presence of clinically significant gastrointestinal disease (for example, clinically active gastroparesis associated with wide glucose fluctuations) in the investigator's opinion

[27] Have excessive insulin resistance defined as having received a total daily dose of insulin >1.5 U/kg at the time of screening

Prior/Concomitant Therapy

[28] Glucocorticoid therapy: Receiving chronic (lasting longer than 14 consecutive days) systemic glucocorticoid therapy (including IV, intramuscular, SC, or oral, but excluding topical, intraocular, intranasal, intra-articular and inhaled preparations) or have received such therapy within 8 weeks immediately prior to screening (Visit 1) with the exception of replacement therapy for adrenal insufficiency

[29] Have been on a treatment regimen that includes regular human insulin or any premixed insulins within 90 days prior to screening (Visit 1)

[30] Have used insulin human inhalation powder (Afrezza®) within 90 days prior to screening (Visit 1)

[31] Have used CSII within the 90 days prior to screening (Visit 1)

[32] Receiving any oral or injectable medication intended for the treatment of diabetes mellitus other than insulins in the 90 days prior to screening (Visit 1)

Prior/Concurrent Clinical Trial Experience

[33] Are currently enrolled in any other clinical trial involving an IP or any other type of medical research judged not to be scientifically or medically compatible with this study

[34] Have participated, within the last 30 days in a clinical trial involving an IP. If the previous IP has a long half-life, 3 months or 5 half-lives (whichever is longer) should have passed

[35] Have previously completed or withdrawn from this study or any other study investigating LY900014 after receiving at least 1 dose of the IP

Other Exclusions

[36] Are investigator site personnel directly affiliated with this study and/or their immediate families. Immediate family is defined as a spouse, parent, child, or sibling, whether biological or legally adopted

[37] Are Lilly employees (including employees, temporary contract workers, or designees responsible for the conduct of the study)

[38] Are unable and/or unwilling to provide informed consent, to make themselves available for the duration of the study, to comply with the use of a data collection device to directly record data, or to abide by study procedures

6.3. Lifestyle Restrictions

- Patients should be instructed not to donate blood or blood products during the study.

- Patients should be instructed to avoid major changes in dietary intake or physical activity during the 3 days prior to MMTT.

6.4. Screen Failures

Individuals who do not meet the criteria for participation in this study (screen failure) may not be rescreened. Retests are also not allowed, except for cases when results are not available from the original sample.